ICICLE
DEMOCRATIZING AI

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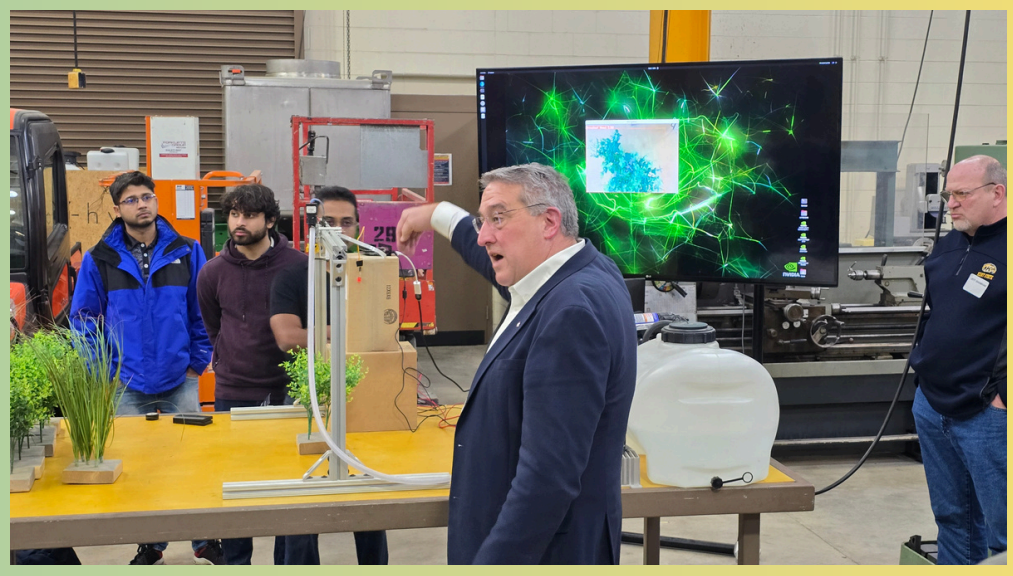
WEBINAR SERIES

AI in Agriculture

Friday, February 27, 2026

11:30 - 12:30 PM Eastern Time

Register here



Abstract: Artificial Intelligence (AI) is rapidly transforming modern agriculture by enabling scalable, data-driven decision-making across the production chain. In this webinar, we showcase how the NSF funded ICICLE AI Institute's **Data-to-Action (DTA) pipeline** supports the development and deployment of intelligent, modular applications for agricultural systems. The session explores end-to-end workflows integrating data ingestion, model training, inference, and interactive decision support through the following demonstrations.

- **Retrieval-Augmented Generation (RAG) chatbot, an AI agent** designed to assist researchers and extension professionals by querying agricultural datasets and ICICLE resources using natural language.
- **Object detection and classification pipelines**, highlighting real-time vision models for identifying crop health conditions and field objects.
- **Precision spot-spraying application** that leverages deep learning for site-specific weed management.

Together, these use cases illustrate ICICLE's approach to making **AI in agriculture accessible, interoperable, and impactful** — fostering collaboration between computer scientists, agricultural engineers, and domain stakeholders to accelerate innovation in sustainable farming systems.

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